

YINGDAN SHI

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EDUCATION

Illinois Institute of Technology

Aug. 2024 - Jun. 2028

Ph.D. in Computer Engineering. Advisor: [Prof. Ren Wang](#)

GPA: 4.0/4.0

Soochow University

Sep. 2021 - Jun. 2024

M.E. in Computer Technology. Advisor: Prof. Jingya Zhou

GPA: 3.7/4.0

Hangzhou Normal University

Sep. 2017 - Jun. 2021

B.E. in Computer Science and Technology.

GPA: 3.8/4.0

RESEARCH INTERESTS

Trustworthy post-training for LLMs and LLM-based agents, with a focus on removing, editing, and controlling learned knowledge and behaviors.

Keywords: LLM Post-training, Safety Alignment, Agent Safety, Self-evolving Agents, Unlearning

SELECTED PUBLICATIONS

(* co-first author)

- [1] [Yingdan Shi](#), Xiang Xu, Kaize Ding, Alfred O. Hero and Ren Wang. “On the Plasticity Collapse in Continual Machine Unlearning.” **ECCV**, 2026. [[Paper](#)][[Code](#)]
- [2] [Yingdan Shi](#) and Ren Wang. “Exploring Nonlinear Pathway in Parameter Space for Machine Unlearning.” **ICML**, 2026. [[Paper](#)] [[Code](#)]
- [3] [Yingdan Shi](#), Sijia Liu, Kaize Ding and Ren Wang. “Tackling Fake Forgetting through Uncertainty Quantification.” **ICML**, 2026. [[Paper](#)] [[Code](#)]
- [4] Jane Downer*, [Yingdan Shi](#)*, Ziyang Liu*, Ren Wang and Binghui Wang. “Watermarking Graph Neural Networks via Explanations for Ownership Protection.” **ICML**, 2026. [[Paper](#)] [[Code](#)]

PREPRINTS

- [1] [Yingdan Shi](#), Ren Wang. “Trajectory Unlearning on LLM-based Agents.” In submission, 2026. [[Paper](#)] [[Code](#)]
- [2] Hyoseo Kim, [Yingdan Shi](#), Ren Wang. “Unmasking Intent Blindness in Generative Engine Optimization.” In submission, 2026.
- [3] Duan Tu, [Yingdan Shi](#), Xu Zhang, Hongkang Li, Ren Wang. “Optimizing over Mode Connectivity for Multi-Task Learning and Machine Unlearning.” In submission, 2026.
- [4] Haoxin Guo, [Yingdan Shi](#), Ren Wang. “Beyond Superficial Metrics: Progressive Layer-wise Expert Unlearning for Deep Erasure in LLMs.” In submission, 2026.

RESEARCH EXPERIENCE

TIML Lab, Illinois Institute of Technology

Aug. 2024 – Present

LLM Agents & Post-training

- **Trajectory Unlearning for LLM Agents** [[Under review, first author](#)]. Formulated *trajectory unlearning*, extending unlearning from factual knowledge to agent *behaviors*. Built agent unlearning benchmarks on ALF-World and WebShop, and proposed **GiRPO**, a GRPO-based on-policy RL algorithm that reduces similarity to forgotten trajectories to **3%** on Qwen2.5-7B-Instruct while preserving task success rate.
- **Self-Unlearning for Self-Evolving Agents** [[Ongoing](#)]. Found that self-evolving agents silently re-acquire unlearned knowledge while evolving; designing self-unlearning inside the evolution loop to prevent re-memorization.
- **Knowledge Entanglement in LLMs** [[Ongoing](#)]. Modeled knowledge as hidden-state *subspaces* (not single vectors) and quantified inter-knowledge entanglement to guide targeted, low-collateral unlearning in LLMs.

Machine Unlearning & Trustworthy ML

- **Plasticity Collapse in Continual Unlearning** [ECCV 2026, first author]. Identified *plasticity collapse*: models gradually fail to forget new requests and re-memorize previously forgotten data. Traced it theoretically to accumulating geometric constraints in parameter space; validated on classification, DDPM, and LLaVA.
- **Nonlinear Unlearning via Mode Connectivity** [ICML 2026, first author]. Showed linear unlearning updates suffer from weight entanglement; proposed a nonlinear pathway along mode-connected curves, a plug-and-play add-on to existing methods that mitigates over- and under-forgetting.
- **Fake Forgetting** [ICML 2026, first author]. Used conformal prediction to show samples “forgotten” under accuracy metrics remain exposed; proposed an uncertainty-aware metric and unlearning framework to close the gap.
- **GNN Watermarking** [ICML 2026, co-first author]. Exposed ownership ambiguity and poisoning risks of backdoor watermarks; proposed an explanation-based watermark requiring no training-data manipulation.

WORK EXPERIENCE

Illinois Institute of Technology	Chicago, IL
Research Assistant	Aug. 2024 - Present
Meituan Inc.	Beijing, China
Business Analyst Internship	May. 2024 - Jul. 2024
Soochow University	Suzhou, China
Research Assistant	Aug. 2021 - Jun. 2024

AWARDS AND HONORS

National Scholarship (2023)
Excellent Master’s Thesis Award (2024)
Outstanding Master’s Award (2024)
Outstanding Undergraduate Award (2021)
First Prize Scholarship (2020, 2021, 2022)

ACADEMIC SERVICE

Conference Reviewer: ICML 2026, SIGIR 2026, NeurIPS 2026, ICLR 2027
Journal Reviewer: IEEE Transactions on Dependable and Secure Computing, World Wide Web, Scientific Reports, Dynamic Games and Applications, Humanities and Social Sciences Communications, Knowledge and Information Systems, Social Network Analysis and Mining